

Series 02 - RStudio

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Statistical test

Part 1: Def

What is a median? A mean?

A median is obtained by adding values and dividing by the number of values. A median is the value in the middle of an ordered list of values.

<https://maths-et-tiques.fr/telech/Tstat1.pdf>

What is a variance? A standard deviation?

A variance indicates how spread out are the values from the mean value. It is calculated as the average of the squared differences between each value and the mean. The standard deviation is the square root of the variance, thus it is expressed in the same unit as the original values.

<https://maths-et-tiques.fr/telech/Tstat1.pdf>

What is a normal distribution?

A normal distribution has a bell shaped, symmetrical curve, 95% of the data lie within two standard derivations of the mean.

Najmi A, Sadasivam B, Ray A. How to choose and interpret a statistical test? An update for budding researchers. J Family Med Prim Care. 2021 Aug;10(8):2763-2767. doi: 10.4103/jfmpc.jfmpc_433_21. Epub 2021 Aug 27. PMID: 34660402; PMCID: PMC8483143.

What is a statistical test?

A statistical test is a method to determine whether there is enough evidence to state that there is a difference in what we want to test.

https://www.itl.nist.gov/div898/handbook/prc/section1/prc13.htm?utm_source=chatgpt.com

When to use a statistical test?

A statistical test is used when we want to determine if the difference observed in the data is real or due to chance.

https://www.itl.nist.gov/div898/handbook/prc/section1/prc13.htm?utm_source=chatgpt.com

What is a parametric test? A non-parametric test?

A parametric test is a type of test used when the data is normally distributed, whereas a non-parametric test is used when the data is not normally distributed.

Najmi A, Sadasivam B, Ray A. How to choose and interpret a statistical test? An update for budding researchers. J Family Med Prim Care. 2021 Aug;10(8):2763-2767. doi: 10.4103/jfmpc.jfmpc_433_21. Epub 2021 Aug 27. PMID: 34660402; PMCID: PMC8483143.

What are the assumptions of parametric tests?

The assumptions of parametric tests are that the data are normally distributed and that the variances are equal between groups.

https://stats.libretexts.org/Bookshelves/Applied_Statistics/Mikes_Biostatistics_Book_%28Dohm%29/13%3A_Assumptions_of_Parametric_Tests?utm_source=chatgpt.com

What are the assumptions of non-parametric tests?

The assumptions of non-parametric tests are that the sampling is random and they assume about the independence or dependence of samples.

https://stats.libretexts.org/Bookshelves/Applied_Statistics/Statistics_for_Research_Students_%28Fein_Gilmour_Machin_and_Liam_Hendry%29/09%3A_Nonparametric_Statistics/9.01%3A_Nonparametric_Definitions?utm_source=chatgpt.com

What is a p-value?

A p-value is a measure which state if a difference is statistically significant. More details to understand better: We start with a null hypothesis that there is no difference between what we want to test. We perform a statistical tests, we obtain a p-value. The p-value is the probability of obtaining a difference at least as large as the one observed, assuming that the null hypothesis is true. Hence, if the p-value is small, the observed result would be unlikely if the null hypothesis was true, so we reject the null hypothesis. Therefore, we state that there is a statistically significant difference (usually when $p < 0.05$).

https://www.itl.nist.gov/div898/handbook/prc/section1/prc131.htm?utm_source=chatgpt.com

What is the risk of error when using a statistical test?

The two types of errors are: - Type I error : when we reject the null hypothesis whereas the null hypothesis was true, - Type II error : when we fail to reject the null hypothesis whereas the null hypothesis was false.

https://online.stat.psu.edu/stat500/Lesson06?utm_source=chatgpt.com

What is the difference between a paired and an unpaired test?

A paired test is used when we test the same group multiple times. An unpaired test is used when testing the observations from two different groups.

https://openbooks.library.unt.edu/quantitative-analysis-exss/chapter/statistical-evaluation-of-differnece/?utm_source=chatgpt.com

Part 2 : Effect of rehabilitation on performance

2.2.1

```
# load the data and improt libraries
library(tidyverse)
```

```
-- Attaching core tidyverse packages ----- tidyverse 2.0.0 --
v dplyr      1.2.1      v readr      2.2.0
v forcats    1.0.1      v stringr    1.6.0
v ggplot2    4.0.3      v tibble     3.3.1
v lubridate  1.9.5      v tidyr      1.3.2
v purrr      1.2.2
```

```
-- Conflicts ----- tidyverse_conflicts() --
```

```
x dplyr::filter() masks stats::filter()
```

```
x dplyr::lag()     masks stats::lag()
```

```
i Use the conflicted package (<http://conflicted.r-lib.org/>) to force all conflicts to become
```

```
library(dplyr)
PrePost <- read.csv("data/PrePost.csv")
head(PrePost)
```

```
  perf  time
1  125 Before
2  130 Before
3  132 Before
4  135 Before
5  136 Before
6  138 Before
```

```
summary(PrePost)
```

```
      perf      time
Min.   :125.0  Length :16
1st Qu.:132.0  N.unique : 2
Median :136.0  N.blank  : 0
Mean   :136.4  Min.nchar: 5
3rd Qu.:140.2  Max.nchar: 6
Max.    :148.0
```

```
glimpse(PrePost)
```

```
Rows: 16
Columns: 2
$ perf <int> 125, 130, 132, 135, 136, 138, 140, 145, 127, 132, 133, 136, 139, ~
$ time <chr> "Before", "Before", "Before", "Before", "Before", "Before", "Before", "Befo~
```

Variable	Description	Unit	Possible values	Expected change
perf	Performance measured during the rehabilitation assessment	Arbitrary units	125-148	Increase after rehabilitation
time	Measurement occasion	Category	Before, After	N/A

Does the treatment have an effect ? To answer, we will compare before and after

```
before <- PrePost$perf[PrePost$time == "Before"]
after  <- PrePost$perf[PrePost$time == "After"]

diff <- after - before # to check the normality of the difference between before and after, v

shapiro.test(diff)
```

Shapiro-Wilk normality test

```
data:  diff
W = 0.89896, p-value = 0.2828
```

What comparison are you going to make?

- Which test will you use? Parametric or non-parametric?

The compare performance before and after rehabilitation on the same participants, we first evaluated the normality of the paired differences, the Shapiro-Wilk test was not significant($p=0.283$), confirming that the normality assumption was satisfied. Because the data are normally distributed and derived from a paired design, a parametric paired t-test is appropriate and was used for the comparison.

```
t.test(after, before, paired = TRUE)
```

Paired t-test

```
data:  after and before
t = 5.4006, df = 7, p-value = 0.001008
alternative hypothesis: true mean difference is not equal to 0
95 percent confidence interval:
 1.405392 3.594608
sample estimates:
mean difference
      2.5
```

Interpretation : A paired t-test showed that performance was significantly higher after rehabilitation than before rehabilitation ($t(7) = 5.40$ from 0, $p = 0.001$). Participants improved by an average of 2.5 units (95% Coincidence Interval [1.41, 3.59]). These results indicate that the rehabilitation program had a significant positive effect on performance.

- Which graph illustrates the effect of the treatment?

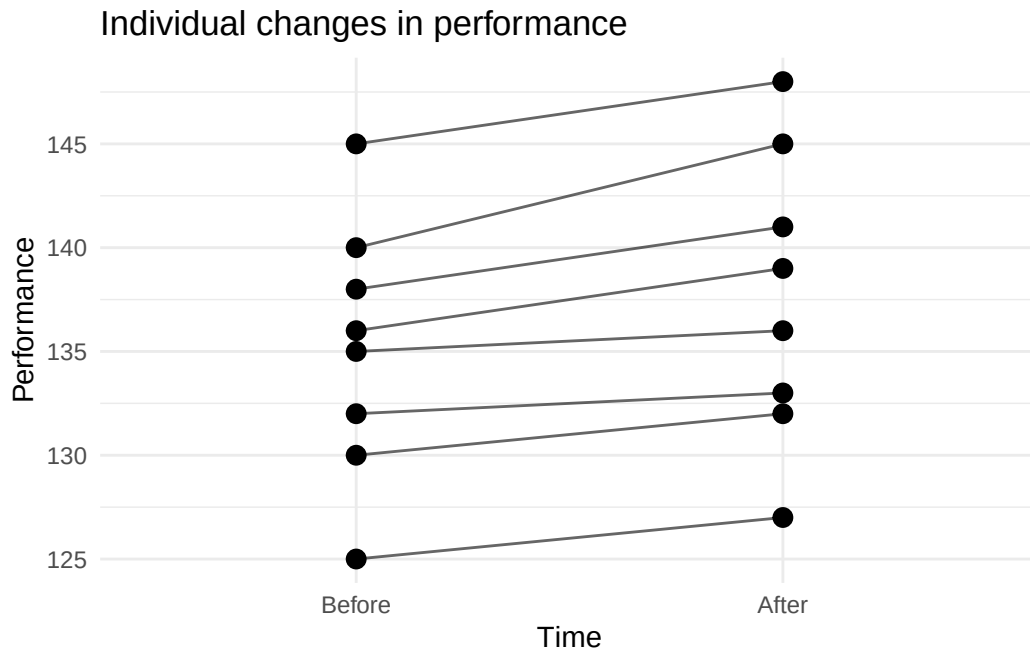
These 2 plots are used to illustrate the effect of the treatment, the boxplot illustrates the distribution of the data and the second one illustrates the paired individual changes between the groups.

```
#(before->after)
PrePost$time <- factor(PrePost$time, levels = c("Before", "After"))

#plot the 1st graph to visualize the distribution and the variation of before and after series
ggplot(
  PrePost,
  aes(x = time, y = perf, fill = time)
) +
  geom_boxplot() +
  theme_minimal() +
  labs(
    x = "Time",
    y = "Performance",
    title = "Effect of rehabilitation on performance"
  )
```



```
PrePost$subject <- rep(1:8, 2)
#plot the 2nd graph to show the individual change due to rehabilitation.
ggplot(
  PrePost,
  aes(x = time, y = perf, group = subject)
) +
  geom_line(alpha = 0.6) +
  geom_point(size = 3) +
  theme_minimal() +
  labs(
    x = "Time",
    y = "Performance",
    title = "Individual changes in performance"
  )
)
```



- What sentence will you write in your thesis to explain your result?

The normality of the differences between pre- and post-treatment performance was first assessed using a Shapiro-Wilk test. The test was not significant ($p = 0.283$), indicating that the normality assumption was satisfied.

Consequently, a paired Student's t-test was performed to compare performance before and after rehabilitation. The analysis revealed a significant increase in performance after treatment ($t(7) = 5.40$, $p = 0.001$).

The mean improvement was 2.5 units (95% CI [1.41, 3.59]), indicating that participants performed significantly better following the rehabilitation program. These results suggest that the rehabilitation intervention had a positive effect on performance.

Limits

A major limitation of this analysis is the small sample size ($n = 8$). The results should therefore be interpreted with caution, as a larger sample would provide more precise and reliable estimates of the treatment effect.

Also, the study did not include a control group. Consequently, it is not possible to determine whether the observed improvement was exclusively due to the rehabilitation program or if other factors occurred.

Testing some stereotypes

```
library(tidyverse)

snore <- read.table(
  "data/snore.txt",
  header = TRUE
)

glimpse(snore)
```

Rows: 100

Columns: 7

```
$ age      <int> 47, 56, 46, 70, 51, 46, 40, 46, 49, 39, 37, 41, 56, 62, 50, 33, ~
$ weight   <int> 71, 58, 116, 96, 91, 98, 112, 77, 76, 119, 99, 117, 83, 68, 108~
$ height   <int> 158, 164, 208, 186, 195, 188, 193, 165, 164, 196, 189, 194, 171~
$ alcohol  <int> 0, 7, 3, 3, 2, 0, 5, 0, 0, 3, 4, 8, 2, 4, 10, 0, 1, 10, 0, 0, 8~
$ sex      <chr> "H", "H", "H", "H", "H", "F", "H", "F", "H", "H", "H", "H", "H", "H"~
$ snore    <chr> "N", "O", "N", "N", "O", "N", "N", "N", "N", "N", "N", "N", "N", "N"~
$ tabac    <chr> "O", "N", "O", "O", "O", "N", "O", "N", "N", "N", "O", "O", "O", "O"~
```

```
summary(snore)
```

age		weight		height		alcohol	
Min.	:23.00	Min.	: 42.00	Min.	:158.0	Min.	: 0.00
1st Qu.	:43.00	1st Qu.	: 77.00	1st Qu.	:166.0	1st Qu.	: 0.00
Median	:52.00	Median	: 95.00	Median	:186.0	Median	: 2.00
Mean	:52.27	Mean	: 90.41	Mean	:181.1	Mean	: 2.95
3rd Qu.	:62.25	3rd Qu.	:107.00	3rd Qu.	:194.0	3rd Qu.	: 4.25
Max.	:74.00	Max.	:120.00	Max.	:208.0	Max.	:15.00

sex		snore		tabac	
Length	:100	Length	:100	Length	:100
N.unique	: 2	N.unique	: 2	N.unique	: 2
N.blank	: 0	N.blank	: 0	N.blank	: 0
Min.nchar	: 1	Min.nchar	: 1	Min.nchar	: 1
Max.nchar	: 1	Max.nchar	: 1	Max.nchar	: 1

The analysis

```
snore$snore <- factor(
  snore$snore,
  levels = c("N","O"),
  labels = c("No","Yes")
)

snore$sex <- factor(
  snore$sex,
  levels = c("F","H"),
  labels = c("Female","Male")
)

snore$tabac <- factor(
  snore$tabac,
  levels = c("N","O"),
  labels = c("No","Yes")
)
```

Are snorers fatter?

```
#Checking if the series of data are follow the normality hypothesis

shapiro.test(
  snore$weight[snore$snore=="Yes"]
)
```

Shapiro-Wilk normality test

```
data:  snore$weight[snore$snore == "Yes"]
W = 0.92744, p-value = 0.0235
```

```
shapiro.test(
  snore$weight[snore$snore=="No"]
)
```

Shapiro-Wilk normality test

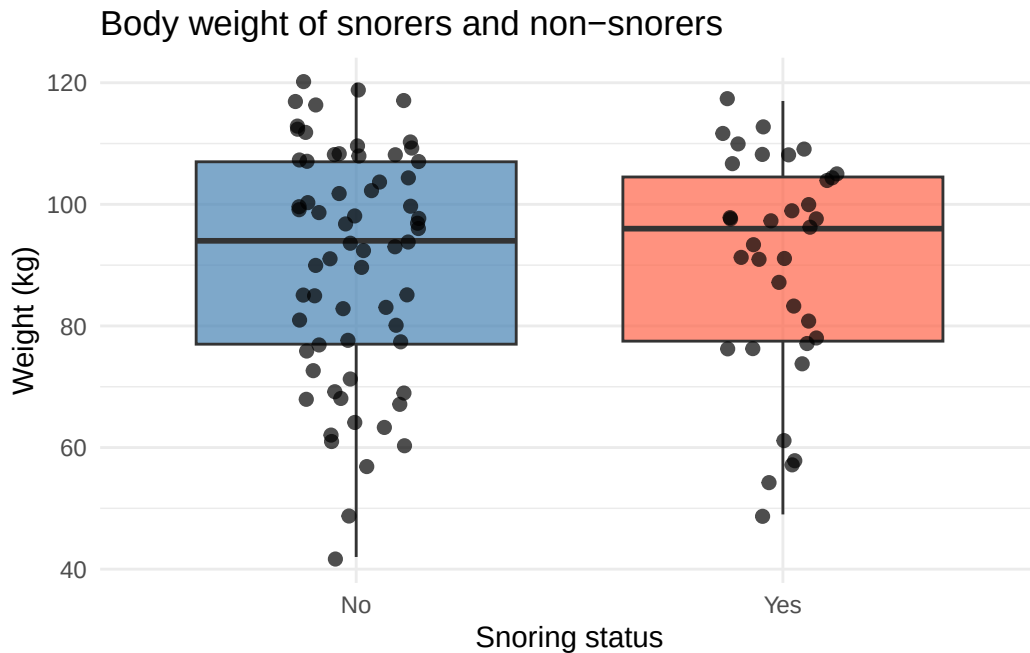
```
data: snore$weight[snore$snore == "No"]  
W = 0.95527, p-value = 0.01955
```

Interpretation

The Shapiro-Wilk test revealed that weight was not normally distributed in either group (snorers: $p = 0.024$; non-snorers: $p = 0.020$).

Therefore, the assumption of normality was not satisfied then a non-parametric test was selected for the comparison.

```
ggplot(  
  snore,  
  aes(x = snore, y = weight, fill = snore)  
) +  
  geom_boxplot(alpha = 0.7, outlier.shape = NA) +  
  geom_jitter(  
    width = 0.15,  
    size = 2,  
    alpha = 0.7,  
    color = "black"  
  ) +  
  scale_fill_manual(  
    values = c(  
      "No" = "steelblue",  
      "Yes" = "tomato"  
    )  
  ) +  
  labs(  
    title = "Body weight of snorers and non-snorers",  
    x = "Snoring status",  
    y = "Weight (kg)"  
  ) +  
  theme_minimal() +  
  theme(  
    legend.position = "none"  
  )  
)
```



Discussions : The boxplot compares body weight between snorers and non-snorers. Snorers appear to have a slightly higher median body weight than non-snorers. However, the distributions overlap considerably, suggesting that any difference may be modest. The variability of body weight is similar in both groups, and several observations are present outside the central range of the data. A statistical test is therefore required to determine whether the observed difference is statistically significant.

```
wilcox.test(weight ~ snore, data = snore)
```

Wilcoxon rank sum test with continuity correction

data: weight by snore

W = 1154, p-value = 0.9079

alternative hypothesis: true location shift is not equal to 0

```
# compare the values between non-snorer & snorer in table
snore %>%
  group_by(snore) %>%
  summarise(
    mean_weight = mean(weight),
    median_weight = median(weight),
```

```
sd_weight = sd(weight)
)
```

```
# A tibble: 2 x 4
  snore mean_weight median_weight sd_weight
<fct>   <dbl>         <int>     <dbl>
1 No      90.5           94      19.0
2 Yes     90.3           96      18.4
```

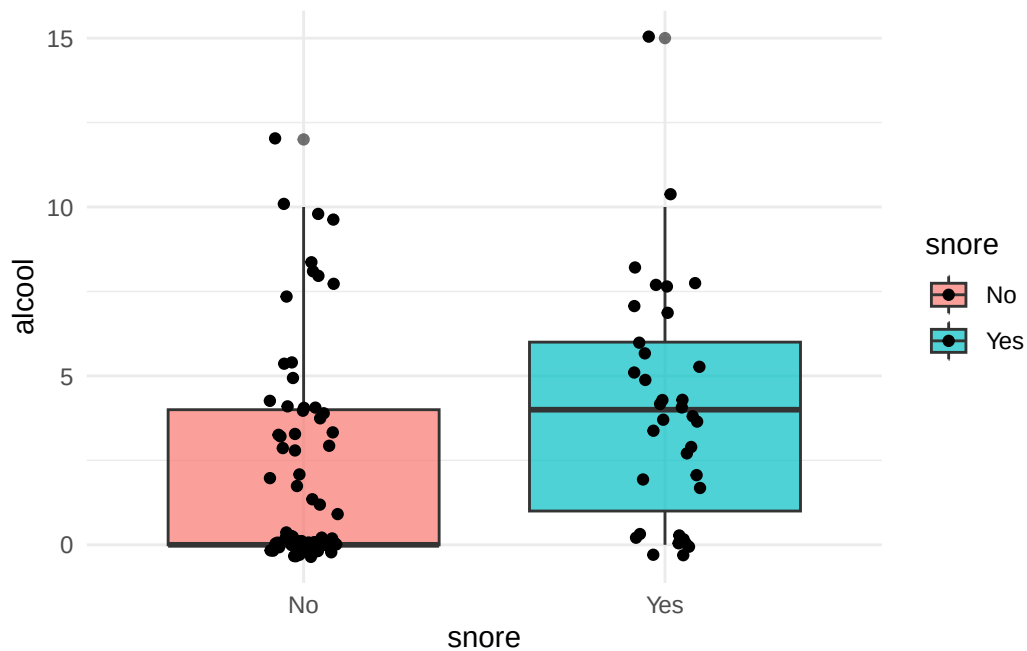
Discussion

To determine whether snorers were heavier than non-snorers, the normality of weight distributions was assessed using the Shapiro-Wilk test. Since normality was rejected in both groups ($p < 0.05$), a Mann-Whitney-Wilcoxon test was performed.

The analysis revealed no significant difference in body weight between snorers and non-snorers ($W = 1154$, $p = 0.908$). The graph coincide with previous results.

Do snorers drink or smoke more

```
ggplot(
  snore,
  aes(x = snore, y = alcohol, fill = snore)
) +
  geom_boxplot(alpha = 0.7) +
  geom_jitter(width = 0.1) +
  theme_minimal()
```



```
#
shapiro.test(
  snore$alcohol[snore$snore=="Yes"]
)
```

Shapiro-Wilk normality test

```
data:  snore$alcohol[snore$snore == "Yes"]
W = 0.90387, p-value = 0.005032
```

```
shapiro.test(
  snore$alcohol[snore$snore=="No"]
)
```

Shapiro-Wilk normality test

```
data:  snore$alcohol[snore$snore == "No"]
W = 0.76256, p-value = 6.818e-09
```

The Shapiro-Wilk test showed that alcohol consumption was not normally distributed among either snorers ($p = 0.005$) or non-snorers ($p < 0.001$). Therefore, a non-parametric Mann-Whitney-Wilcoxon test was used for further analysis.

And i chose to plot the distribution of alcohol consumption between snorers and non-snorers using a boxplot with jittered points to visualize the data.

```
wilcox.test(  
  alcool ~ snore,  
  data = snore  
)
```

Wilcoxon rank sum test with continuity correction

```
data:  alcool by snore  
W = 788, p-value = 0.008594  
alternative hypothesis: true location shift is not equal to 0
```

The test revealed a significant difference in alcohol consumption between snorers and non-snorers ($W = 788$, $p = 0.009$).

Descriptive statistics showed that snorers consumed more alcohol on average than non-snorers (mean = 4.03 vs 2.37). The median alcohol consumption was also higher among snorers (4 vs 0).

These results suggest that snorers consume significantly more alcohol than non-snorers.

```
#See the value in a table  
table(snore$snore, snore$tabac)
```

	No	Yes
No	21	44
Yes	15	20

```
#Seeing correlation between  
chisq.test(  
  table(snore$snore, snore$tabac)  
)
```

Pearson's Chi-squared test with Yates' continuity correction

```
data: table(snore$snore, snore$tabac)
X-squared = 0.68872, df = 1, p-value = 0.4066
```

To investigate whether smoking was associated with snoring, a Chi-square test of independence was performed.

The analysis revealed no significant association between smoking status and snoring ($\chi^2(1) = 0.689$, $p = 0.407$).

Among non-snorers, 67.7% were smokers, whereas 57.1% of snorers were smokers. Although these proportions differed slightly, the difference was not statistically significant.

Therefore, smoking status does not appear to be associated with snoring in the present sample.

Overall, alcohol consumption and smoking status were investigated as potential factors associated with snoring.

A Mann-Whitney-Wilcoxon test revealed that snorers consumed significantly more alcohol than non-snorers ($W = 788$, $p = 0.009$). Snorers showed both a higher mean alcohol consumption (4.03 vs 2.37) and a higher median alcohol consumption (4 vs 0).

In contrast, a Chi-square test indicated no significant association between smoking status and snoring ($\chi^2(1) = 0.689$, $p = 0.407$).

These findings suggest that alcohol consumption is associated with snoring, whereas smoking does not appear to be associated with snoring in this sample.

Limit of the study : Exclusion of medical pathology among the population.

```
# Library
library(tidyverse)
library(dplyr)

# Import data
snore <- read.table("data/snore.txt", header = TRUE)

# Verify
head(snore)
```

	age	weight	height	alcohol	sex	snore	tabac
1	47	71	158	0	H	N	0
2	56	58	164	7	H	O	N
3	46	116	208	3	H	N	0

4	70	96	186	3	H	N	O
5	51	91	195	2	H	O	O
6	46	98	188	0	F	N	N

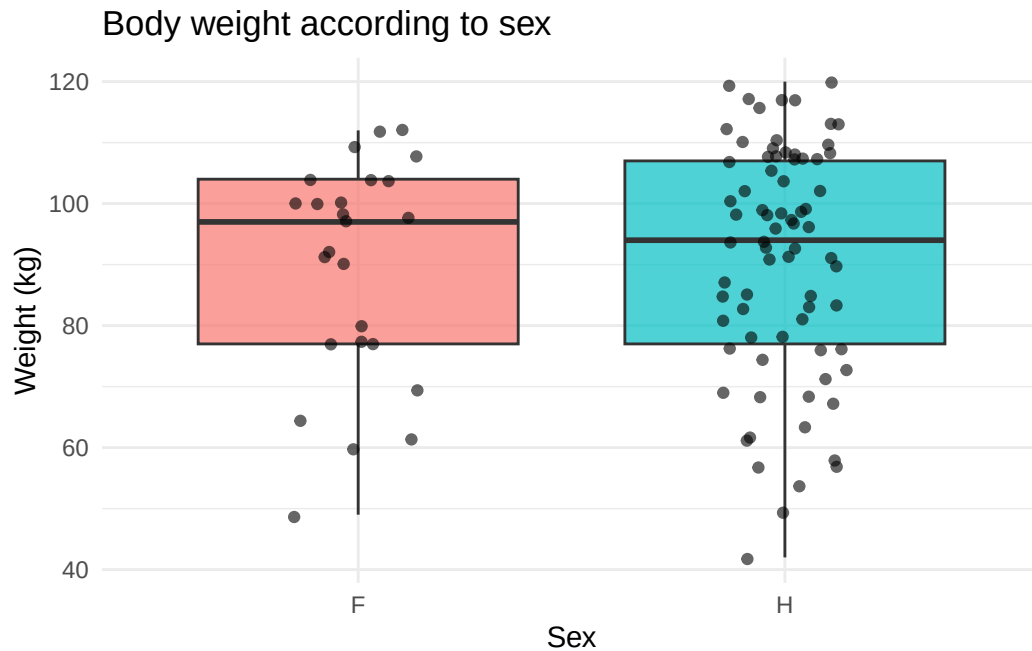
```
str(snore)
```

```
'data.frame': 100 obs. of 7 variables:
 $ age : int 47 56 46 70 51 46 40 46 49 39 ...
 $ weight: int 71 58 116 96 91 98 112 77 76 119 ...
 $ height: int 158 164 208 186 195 188 193 165 164 196 ...
 $ alcool: int 0 7 3 3 2 0 5 0 0 3 ...
 $ sex : chr "H" "H" "H" "H" ...
 $ snore : chr "N" "O" "N" "N" ...
 $ tabac : chr "O" "N" "O" "O" ...
```

Are men fatter ?

To answer this question, we will first plot the data.

```
#
ggplot(
  snore,
  aes(x = sex, y = weight, fill = sex)#fill #: we use fill to difference the two groups
) +
  geom_boxplot(outlier.shape = NA, alpha = 0.7) +
  geom_jitter(#some people same weight so shift a bit the points for more visability
    width = 0.15,
    alpha = 0.6
  ) +
  theme_minimal() +
  theme(
    legend.position = "none"#because sex is already indicated on the x axis
  ) +
  labs(
    title = "Body weight according to sex",
    x = "Sex",
    y = "Weight (kg)"
  )
```



Now we will test if the data are normally distributed or not.

```
shapiro.test(
  snore$weight[snore$sex == "H"]
)
```

Shapiro-Wilk normality test

```
data: snore$weight[snore$sex == "H"]
W = 0.95406, p-value = 0.008379
```

```
shapiro.test(
  snore$weight[snore$sex == "F"]
)
```

Shapiro-Wilk normality test

```
data: snore$weight[snore$sex == "F"]
W = 0.9146, p-value = 0.03865
```

The Shapiro-Wilk test showed that weight was not normally distributed in either group, with $p=0.008$ for men and $p=0.039$ for women.

Therefore, we will use Mann-Whitney-Wilcoxon because we want to compare two independent groups and because the data are not normally distributed.

```
wilcox.test(  
  weight ~ sex, #compare weight according to sex  
  data = snore  
)
```

Wilcoxon rank sum test with continuity correction

data: weight by sex

W = 893.5, p-value = 0.729

alternative hypothesis: true location shift is not equal to 0

```
snore %>% # %>% means take this result and give it to the next step  
  group_by(sex) %>% #for each sex:  
  summarise(  
    mean_weight = mean(weight),  
    median_weight = median(weight),  
    sd_weight = sd(weight)  
  )
```

```
# A tibble: 2 x 4  
  sex    mean_weight median_weight sd_weight  
  <chr>      <dbl>         <int>      <dbl>  
1 F          89.3           97        18.1  
2 H          90.8           94        19.1
```

The Mann-Whitney-Wilcoxon showed no significant difference in body weight between men and women ($W=893.5$, $p=0.729$).

The mean body weight was 90.77kg for men and 89.32kg for women.

Therefore, we found no statistically significant evidence that men were heavier than women in our sample.

Do women smoke less?

First, we will create a table and a plot to have an idea of the data.

```
#showing the % of smoking or not according to gender in a table
# N: Non smoker, O: Smoker
print("Table of smoker according to gender in percentage():")
```

```
[1] "Table of smoker according to gender in percentage():"
```

```
smoking_table <- prop.table(
  table(snore$sex, snore$tabac),
  margin = 1 #calculate the proportion by line
) * 100 #to read it as percentage

rownames(smoking_table) <- c("Female", "Male")
colnames(smoking_table) <- c("No", "Yes")

print(smoking_table)
```

	No	Yes
Female	60	40
Male	28	72

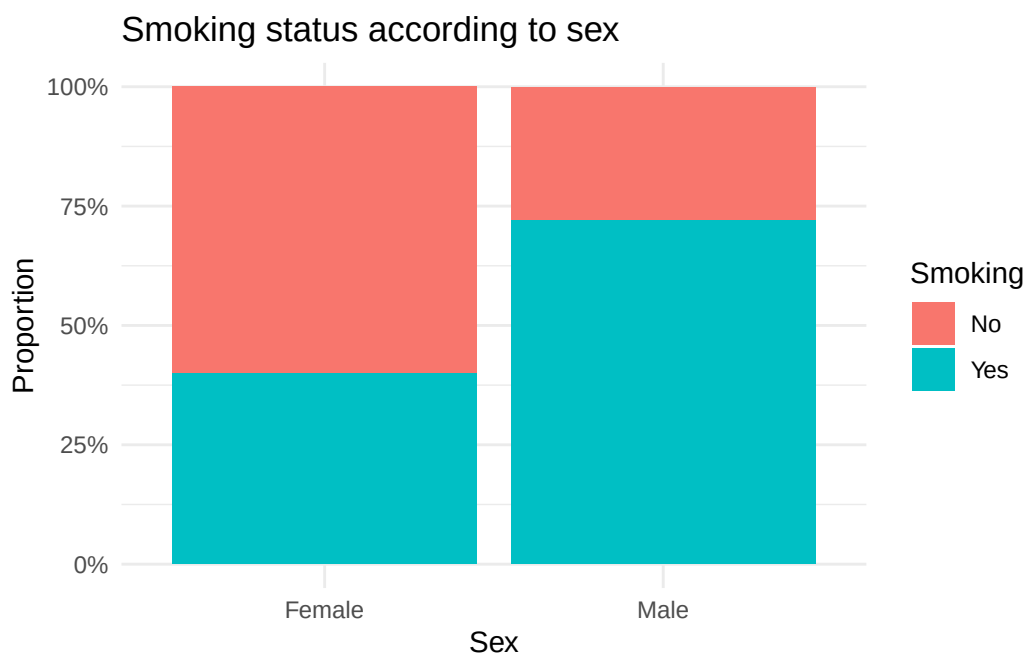
```
#graph
ggplot(
  snore,
  aes(
    x = sex,
    fill = tabac
  )
) +
geom_bar(position = "fill") +
scale_x_discrete(
  labels = c(
    "F" = "Female",
    "H" = "Male"
  )
) +
scale_fill_discrete(
```

```

labels = c(
  "N" = "No",
  "O" = "Yes"
)
) +

scale_y_continuous(labels = scales::percent) +
theme_minimal() +
labs(
  title = "Smoking status according to sex",
  x = "Sex",
  y = "Proportion",
  fill = "Smoking"
)

```



Because these two variables are categorical variables we use Chi-square test of independence to test whether sex and tabac are independent.

```
chisq.test(table(snore$sex, snore$tabac))
```

Pearson's Chi-squared test with Yates' continuity correction

```
data: table(snore$sex, snore$tabac)
X-squared = 7.0023, df = 1, p-value = 0.00814
```

The chi-square test showed a significant association between sex and smoking status ($\chi^2 = 7.0023$, $df = 1$, $p = 0.00814$). In our sample, 40% of women were smokers compared with 72% of men. Therefore, the proportion of smokers was significantly lower among women than among men.

From a more general perspective: Are there any correlations between variables?

We are interested in the quantitative variables:

- Age
- Weight
- Height
- Alcohol

Therefore, we use Spearman's correlation to evaluate the correlations between these variables.

```
num_data <- snore %>%
  select(age, weight, height, alcohol)
#To detect correlation between numerical variables
cor(
  num_data,
  method = "spearman",
  use = "complete.obs"
)
```

	age	weight	height	alcohol
age	1.000000000	-0.01455125	0.008072461	-0.10748796
weight	-0.014551254	1.000000000	0.930383489	0.08568687
height	0.008072461	0.93038349	1.000000000	0.12539769
alcohol	-0.107487956	0.08568687	0.125397694	1.000000000

The correlation analysis revealed very few relationships between the quantitative variables.

It showed one strong positive correlation between height and weight, with a Spearman correlation coefficient of 0.93. This means that taller individuals tended to be heavier and, heavier individuals tended to be taller. The other correlations were weak and close to zero.

Conclusion

The correlation analysis revealed a strong positive association between height and weight ($r = 0.93$), indicating that taller individuals generally tended to be heavier. In contrast, age and alcohol consumption showed only weak or negligible correlations with the other quantitative variables. Overall, height was the only variable strongly associated with another measure in this dataset.

A limitation of this study is the relatively small sample size, which may reduce the generalizability of the results. Future studies should include larger and more diverse populations to better understand the factors associated with snoring.